

RAMATENKI SRINIVASA RAO

AI Engineer

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SUMMARY

AI Engineer with a strong Computer Science foundation and expertise in building intelligent, scalable backend systems. Proficient in integrating Large Language Models (LLMs) and Vector Databases to create advanced AI solutions. Skilled in Python frameworks (FastAPI/Django) and asynchronous programming to optimize AI model deployment and reduce latency in production environments.

SKILLS

AI & Machine Learning: LLM Integration, Vector Databases, Semantic Search, Embedding Generation, Pydantic V2

Languages & Web: Python (PEP 8), SQL, JavaScript, HTML, CSS

Frameworks: FastAPI, Django, Django Rest Framework

Infrastructure & Tools: PostgreSQL, Redis, SQLAlchemy, Docker, AWS, Celery, Git, Pytest

PROJECTS

AI-POWERED SEMANTIC SEARCH API

- Engineered a semantic search engine using FastAPI and Vector Databases to enable "search by meaning" rather than just keywords.
- Optimized performance by implementing Async for non-blocking embedding generation, allowing the system to process 500+ document queries concurrently.
- Reduced redundant AI model calls by 30% using Redis caching, significantly lowering potential API latency and compute costs.
- Utilized Pydantic V2 for strict data validation and SQLAlchemy 2.0 (Async) for efficient database transactions.
- Engineered an asynchronous monitoring service using FastAPI and Async, leveraging non-blocking I/O to handle 1k+ concurrent pings.
- Implemented WebSockets for real-time telemetry, ensuring sub 100ms state updates for AI-driven frontend dashboards.

MULTI-TENANT B2B PLATFORM

- Developed a multi-tenant backend using Django DRF, implementing custom Middleware for tenant-aware data routing and secure isolation.
- Optimized database interaction using Django QuerySet optimization, reducing DB hits by 60% for high-traffic AI applications.
- Integrated Celery and Redis for asynchronous task processing, ensuring the main event loop remains non-blocking for high-priority requests.

EDUCATION

UNIVERSITY COLLEGE OF ENGINEERING(KU) , KOTHAGUDEM
B.Tech in Computer Science and Engineering
CGPA: 8.00

2022 – 2026